🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.7

Ordered Pairs in All Four Quadrants

Lesson 7-8 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes (Level 2 Standard)

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can plot ordered pairs in all four quadrants of the coordinate plane.
Language Objective: I can explain my work using the words quadrant, negative coordinate, reflection, and axis.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Ordered Pairs in All Four Quadrants | Quadrant | Negative coordinate | Reflection | Axis | Integer

👆 *Tap any word to see what it means and a picture.*

1. ________ — Points written as (x, y) whose positive or negative coordinates place them in one of four quadrants.

2. One of the four regions of the coordinate plane, numbered I to IV, is a ________.

3. A coordinate less than zero, which places a point left of or below the origin, is a ________.

4. A flip of a point over an axis is a ________.

5. One of the two number lines, the x-axis or y-axis, that form the grid is an ________.

6 A whole number or its opposite is an ______ .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How do four-quadrant coordinates help a city map locations in every direction from the center?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Ordered Pairs in All Four Quadrants** — Points written as (x, y) whose positive or negative coordinates place them in one of four quadrants.
  Pares ordenados en los cuatro cuadrantes — Puntos escritos como (x, y) cuyas coordenadas positivas o negativas los ubican en uno de cuatro cuadrantes.

- [ ] **Quadrant** — One of the four parts of a coordinate grid, numbered I to IV.
  Cuadrante — Una de las cuatro partes de una cuadrícula de coordenadas, numeradas I a IV.

- [ ] **Negative coordinate** — A coordinate less than zero. It means left or below the center point.
  Coordenada negativa — Una coordenada menor que cero. Significa a la izquierda o debajo del centro.

- [ ] **Reflection** — A flipped, mirror image across a line.
  Reflexión — Una imagen reflejada al otro lado de una línea.

- [ ] **Axis** — A line on the grid. The x-axis goes across; the y-axis goes up.
  Eje — Una línea en la cuadrícula. El eje x va de lado; el eje y va hacia arriba.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The library is in Quadrant ____ because its x is ____ and its y is ____. The park is in Quadrant ____ because ____. Four-quadrant coordinates help because ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The signs of the coordinates tell me the quadrant because ____.

Evidence: Student matches sign combinations to quadrants (e.g., (+,+) is I, (-,+) is II).

Reasoning: This evidence connects the definition of ordered pairs in all four quadrants to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The library is in Quadrant ____ because its x is ____ and its y is ____. The park is in Quadrant ____ because ____. Four-quadrant coordinates help because ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Ordered Pairs in All Four Quadrants
2. **Notes 2:** Quadrant
3. **Notes 3:** Negative coordinate
4. **Notes 4:** Reflection
5. **Notes 5:** Axis
6. **Notes 6:** Integer
7. **Watch for this mistake:** *A common mistake in Ordered Pairs in All Four Quadrants is matching each sign to the wrong axis — for example, seeing (-2, 5) and thinking "the x is negative so I move DOWN, and the y is positive so I move RIGHT," which lands the point in Quadrant IV. But the x-coordinate always controls left/right, never up/down: negative x means LEFT (not down), and positive y means UP (not right). So (-2, 5) is actually left 2, up 5 — Quadrant II. Before you submit, check that you used the x-coordinate for left/right and the y-coordinate for up/down, and never swapped them.*
8. **Try It 1:** C. Quadrant III — *A negative x means left and a negative y means down, so (-5, -2) is in Quadrant III, where both coordinates are negative.*
9. **Try It 2:** B. (-, +) — *Quadrant II is up and to the left of the origin, so x is negative and y is positive: (-, +).*